CS 70 Discrete Mathematics and Probability Theory
Fall 2025 Course Notes Note 1

1 Propositional Logic

In order to be fluent in working with mathematical statements, you need to understand the basic framework of the language of mathematics. This first lecture, we will start by learning about what logical forms mathematical theorems may take, and how to manipulate those forms to make them easier to prove. In the next few lectures, we will learn several different methods of proving things.

Our first building block is the notion of a **proposition**, which is simply a statement which is either true or false.

These statements are all propositions:

(1) $\sqrt{3}$ is irrational.

(2) $1+1=5$.

(3) Julius Caesar had 2 eggs for breakfast on his 10th birthday.

These statements are clearly not propositions:

(4) $2+2$.

(5) $x^2+3x=5$. [What is x?]

These statements aren't propositions either (although some books say they are). Propositions should not include fuzzy terms.

(6) Arnold Schwarzenegger often eats broccoli. [What is "often?"]

(7) Henry VIII was unpopular. [What is "unpopular?"]

Propositions may be joined together to form more complex statements. Let P, Q, and R be variables representing propositions (for example, P could stand for "3 is odd"). The simplest way of joining these propositions together is to use the connectives "and", "or" and "not."

(1) **Conjunction**: $P \wedge Q$ ("P and Q"). True only when both P and Q are true.

(2) **Disjunction**: $P \vee Q$ ("P or Q"). True when at least one of P and Q is true.

(3) **Negation**: $\neg P$ ("not P"). True when P is false.

Statements like these, with variables, are called *propositional forms*.

A fundamental principle known as the **law of the excluded middle** says that, for any proposition P, either P is true or $\neg P$ is true (but not both). Thus $P \vee \neg P$ is always true, regardless of the truth value of P. A propositional form that is always true regardless of the truth values of its variables is called a *tautology*. Conversely, a statement such as $P \wedge \neg P$, which is always false, is called a *contradiction*.

CS 70 离散数学与概率论 2025 年秋季课程笔记

注 1

1 命题逻辑

为了能够熟练地运用数学命题，你需要理解数学语言的基本框架。在第一讲中，我们将首先学习数学定理可以采取哪些逻辑形式，以及如何对这些形式进行变换，使其更易于证明。在接下来的几讲中，我们将学习几种不同的证明方法。

我们的第一个基本组成部分是命题的概念，命题就是一个真或假的陈述。

这些陈述都是命题：

(1) $\sqrt{3}$ 是无理数。

（2）1 + 1 = 5。

（3）尤利乌斯・凯撒 10 岁生日那天早餐吃了 2 个鸡蛋。

这些陈述显然不是命题：

（4）2 + 2。

（5）x + 3x = 5。[x 是多少？]

这些陈述也不是命题（尽管有些书称它们是）。命题不应该包含模糊术语。

（6）阿诺・施瓦辛格经常吃西兰花。[“经常”指的是什么？]

（7）亨利八世不受欢迎。[“不受欢迎”是什么意思？]

命题可以组合成更复杂的陈述。设 P、Q 和 R 为代表命题的变量（例如，P 可以代表“3 是奇数”）。连接这些命题的最简单方法是使用连接词“和”、“或”和“非”。

（1）合取式：P ∧ Q（“P 且 Q”）。仅当 P 和 Q 都为真时才为真。

（2）析取：P ∨ Q（“P 或 Q”）。当 P 和 Q 中至少有一个为真时，P ∨ Q 为真。

（3）否定：¬P（“非 P”）。当 P 为假时，¬P 为真。

像这样带有变量的陈述称为命题形式。

排中律这一基本原理指出，对于任何命题 P，要么 P 为真，要么 ¬P 为真（二者不可兼得）。因此，P ∨ ¬P 始终为真，无论 P 的真值如何。无论其变量的真值如何，始终为真的命题形式称为重言式。

反之，像 P ∧ ¬P 这样总是错误的陈述称为矛盾。

Concept check! If we let P stand for the proposition "3 is odd", Q stand for "4 is odd", and R for "$4+5=49$", what are the values of $P \wedge R$, $P \vee R$ and $\neg Q$?

A useful tool for describing the possible values of a propositional form is a **truth table**. Truth tables are the same as function tables: you list all possible input values for the variables, and then list the outputs given those inputs. (The order does not matter.)

Here are truth tables for conjunction and negation:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	$\neg P$
T	F
F	T

Concept check! Write down the truth table for disjunction (OR).

The most important and subtle propositional form is an **implication**:

(4) **Implication**: $P \implies Q$ ("P implies Q"). This is the same as "If P, then Q."

Here, P is called the *hypothesis* of the implication, and Q is the *conclusion*.[1]

We encounter implications frequently in everyday life; here are a couple of examples:

> If you stand in the rain, then you'll get wet.
>
> If you passed the class, you received a certificate.

An implication $P \implies Q$ is false only when P is true and Q is false. For example, the first statement above would be false only if you stood in the rain but didn't get wet. The second statement would be false only if you passed the class but didn't receive a certificate.

Here is the truth table for $P \implies Q$ (along with an extra column that we'll explain in a moment):

P	Q	$P \implies Q$	$\neg P \vee Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Note that $P \implies Q$ is always true when P is false. This means that many statements that sound nonsensical in English are true, mathematically speaking. Examples are statements like: "If pigs can fly, then horses can read" or "If 14 is odd then $1+2=18$." (These are statements that we never make in everyday life, but are perfectly natural in mathematics.) When an implication is stupidly true because the hypothesis is false, we say that it is *vacuously true*.

Note also that $P \implies Q$ is **logically equivalent** to $\neg P \vee Q$, as can be seen by comparing the last two columns of the above truth table: for all possible truth values of P and Q, $P \implies Q$ and $\neg P \vee Q$ take the same values

[1] P is also sometimes called the *antecedent* and Q the *consequent*.

概念检查！如果我们用 P 表示命题“3 是奇数”，Q 表示命题“4 是奇数”，R 表示命题“4 + 5 = 49”，那么 P ∧ R、P ∨ R 和 ¬Q 的值分别是多少？

描述命题形式所有可能取值的有效工具是真值表。真值表与函数表类似：列出变量的所有可能输入值，然后列出给定这些输入值对应的输出值。 顺序无关紧要。）

以下是合取和否定的真值表：

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

P	¬P
T	F
F	T

概念检查！写出析取（或）的真值表。

最重要、最精妙的命题形式是蕴涵：

（4）蕴含：P =⇒ Q（“P 蕴含 Q”）。这与“如果 P，则 Q”相同。

这里，P 被称为蕴涵式的假设，Q 被称为结论。

我们在日常生活中经常会遇到隐含意义；以下是两个例子：

如果你站在雨里，就会被淋湿。

如果你通过了课程，就会获得证书。

蕴含式 P ⇒ Q 仅当 P 为真且 Q 为假时才为假。例如，上面的第一个命题仅当你站在雨中却没被淋湿时才为假。第二个命题仅当你通过了考试却没拿到证书时才为假。

以下是 P = ⇒ Q 的真值表（以及稍后我们将解释的额外一列）：

P	Q	P =⇒ Q	¬P ∨ Q
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

注意，当 P 为假时，P ⇒ Q 总是为真。这意味着许多在英语中听起来荒谬的陈述，从数学角度来说却是正确的。例如： “如果猪会飞，那么马会读书”或“如果 14 是奇数，那么 1 + 2 = 18”。 这些陈述我们在日常生活中绝不会说，但在数学中却完全自然。）当一个推论因为其假设为假而显得荒谬地正确时，我们称之为空洞的正确。

另请注意，P ⇒ Q 在逻辑上等价于 ¬P ∨ Q，这可以通过比较上述真值表的最后两列看出：对于 P 和 Q 的所有可能真值，P ⇒ Q 和 ¬P ∨ Q 都取相同的值。

1P 有时也被称为前件，Q 被称为后件。

(i.e., they have the same truth table). We write this as $(P \implies Q) \equiv (\neg P \vee Q)$.

$P \implies Q$ is the most common form mathematical theorems take. Here are some of the different ways of saying it:

(1) if P, then Q;

(2) Q if P;

(3) P only if Q;

(4) P is sufficient for Q;

(5) Q is necessary for P;

(6) Q unless not P.

If both $P \implies Q$ and $Q \implies P$ are true, then we say "P if and only if Q" (abbreviated "P iff Q"). Formally, we write $P \iff Q$. Note that $P \iff Q$ is true only when P and Q have the same truth values (both true or both false).

For example, if we let P be "3 is odd," Q be "4 is odd," and R be "6 is even," then the following are all true: $P \implies R$, $Q \implies P$ (vacuously), and $R \implies P$. Because $P \implies R$ and $R \implies P$, we also see that $P \iff R$ is true.

Given an implication $P \implies Q$, we can also define its

(a) **Contrapositive**: $\neg Q \implies \neg P$

(b) **Converse**: $Q \implies P$

The contrapositive of "If you passed the class, you received a certificate" is "If you did not get a certificate, you did not pass the class." The converse is "If you got a certificate, you passed the class." Does the contrapositive say the same thing as the original statement? Does the converse?

Let's look at the truth tables:

P	Q	$\neg P$	$\neg Q$	$P \implies Q$	$Q \implies P$	$\neg Q \implies \neg P$	$P \iff Q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

Note that $P \implies Q$ and its contrapositive have the *same* truth values everywhere in their truth tables, so they are logically equivalent: $(P \implies Q) \equiv (\neg Q \implies \neg P)$. Many students confuse the contrapositive with the converse: note that $P \implies Q$ and $\neg Q \implies \neg P$ are logically equivalent, but $P \implies Q$ and $Q \implies P$ are not!

When two propositional forms are logically equivalent, we can think of them as "meaning the same thing." We will see in the next lecture how useful this can be for proving theorems.

2 Quantifiers

The mathematical statements you'll see in practice will not be made up of simple propositions like "3 is odd." Instead you'll see statements like:

(1) For all natural numbers n, $n^2 + n + 41$ is prime.

（即，它们具有相同的真值表）。我们将其记为（P =⇒ Q）≡（¬P ∨ Q）。

P = ⇒ Q 是数学定理最常见的形式。以下是一些不同的表达方式：

（1）如果 P，则 Q；

（2）Q 如果 P；

（3）P 仅当 Q 时成立；

（4）P 足以满足 Q；

（5）Q 是 P 的必要条件；

（6）除非不是 P，否则为 Q。

如果 P ⇐ ⇒ Q 和 Q ⇐ ⇒ P 都为真，我们就说“P 当且仅当 Q” 简写为“P iff Q”）。形式上，我们记作 P ⇐ ⇒ Q。注意，P ⇐ ⇒ Q 仅当 P 和 Q 的真值相同时才为真（要么都为真，要么都为假）。

例如，如果我们令 P 为“3 是奇数”，Q 为“4 是奇数”，R 为“6 是偶数”，那么以下所有命题都成立：P =⇒ R，Q =⇒ P（显然成立），R =⇒ P。因为 P =⇒ R 且 R =⇒ P，所以我们也可以看出 P ⇐⇒ R 成立。

给定蕴涵式 P =⇒ Q，我们也可以定义它的

（a）逆否命题：¬Q =⇒ ¬P

（b）逆命题：Q =⇒ P

“如果你通过了这门课，你就获得了证书”的逆否命题是“如果你没有获得证书，你就没有通过这门课”。它的逆否命题是“如果你获得了证书，你就通过了这门课”。逆否命题和原命题表达的意思一样吗？它的逆否命题表达的意思一样吗？

我们来看一下真值表：

PQ	¬P	¬QP =⇒ QQ	=⇒ P	¬Q =⇒	¬PP	⇐⇒	QTTFFTTTTTFFTFTFFFTTF
TFTFFFTTTTTTT							

注意，P ⇒ Q 及其逆否命题在真值表中处处具有相同的真值，因此它们逻辑等价：(P ⇒ Q) ≡ (¬Q ⇒ ¬P)。许多学生会将逆否命题与逆命题混淆：请注意，P ⇒ Q 和 ¬Q ⇒ ¬P 逻辑等价，但 P ⇒ Q 和 Q ⇒ P 逻辑不等价！

当两个命题形式在逻辑上等价时，我们可以认为它们“意思相同”。

我们将在下一讲中看到它在证明定理方面有多么有用。

2 量词

你在实践中看到的数学命题并非像“3 是奇数”这样简单的命题，而是会像这样的命题：

（1）对于所有自然数 n，n+n+41 是质数。

(2) If n is an odd integer, so is n^3.

(3) There is an integer k that is both even and odd.

In essence, these statements assert something about lots of simple propositions (even infinitely many!) all at once. For instance, the first statement is asserting that 0^2+0+41 is prime, 1^2+1+41 is prime, and so on. The last statement is saying that, as k ranges over all possible integers, we will find at least one value of k for which the statement is satisfied.

Why are the above three examples considered to be propositions, while earlier we claimed that "$x^2+3x=5$" was not? The reason is that in these examples, there is an underlying "universe" that we are working in, and the statements are *quantified* over that universe. To express these statements mathematically we need two **quantifiers**: The *universal quantifier* $\forall$ ("for all") and the *existential quantifier* $\exists$ ("there exists"). Examples:

(1) "Some mammals lay eggs." Mathematically, "some" means "at least one," so the statement is saying "There exists a mammal x such that x lays eggs." If we let our universe U be the set of mammals, then we can write: $(\exists x \in U)(x \text{ lays eggs})$. (Sometimes, when the universe is clear, we omit U and simply write $\exists x(x \text{ lays eggs})$.)

(2) "For all natural numbers n, n^2+n+41 is prime," can be expressed by taking our universe to be the set of natural numbers, denoted as $\mathbb{N}$: $(\forall n \in \mathbb{N})(n^2+n+41 \text{ is prime})$.

We refer to a statement which refers to a variable as a *predicate* or as a *propositional formula* when replacing the variable with a value makes the statement either true or false. For example, the statement "n^2+n+41 is prime" is a predicate or propositional formula with variable n. The value of the statement on natural numbers, n, is either true or false depending on the value of n. That is, replacing the variable with a value makes the predicate into a proposition. In particular, for $n=1$, we have "43 is prime" which is true. Where for $n=41$, the statement "$41^2+41+41$ is prime" is false as one can factor $41^2+41+41$ into 41×43.

Note that in a *finite* universe, we can express existentially and universally quantified propositions without quantifiers, using disjunctions and conjunctions respectively. For example, if our universe U is $\{1,2,3,4\}$, then $(\exists x \in U)P(x)$ is logically equivalent to $P(1)\vee P(2)\vee P(3)\vee P(4)$, and $(\forall x \in U)P(x)$ is logically equivalent to $P(1)\wedge P(2)\wedge P(3)\wedge P(4)$. However, in an infinite universe, such as the natural numbers, this is not possible.

Concept check! Use quantifiers to express the following two statements: "For all integers x, $2x+1$ is odd", and "There exists an integer between 2 and 4".

Some statements can have multiple quantifiers. As we will see, however, quantifiers do not commute. You can see this just by thinking about English statements. Consider the following (rather gory) example:

> "Every time I ride the subway in New York, somebody gets stabbed."
>
> "There is someone, such that every time I ride the subway in New York, that someone gets stabbed."

The first statement is saying that every time I ride the subway someone gets stabbed, but it could be a different person each time. The second statement is saying something truly horrible: that there is some poor guy Joe with the misfortune that every time I get on the New York subway, there is Joe, getting stabbed again. (Poor Joe will run for his life the second he sees me.)

Mathematically, we are quantifying over two universes: $T=\{\text{times when I ride on the subway}\}$ and $P=\{\text{people}\}$. The first statement can be written: $(\forall t \in T)(\exists p \in P)(p \text{ gets stabbed at time } t)$. The second

（2）如果 n 是奇数，那么 n 也是奇数。

（3）存在一个整数 k，它既是偶数又是奇数。

本质上，这些语句同时断言了许多简单命题（甚至是无穷多个命题！）的性质。例如，第一个语句断言 0 + 0 + 41 是素数，1 + 1 + 41 也是素数，以此类推。最后一个语句则表明，当 k 取遍所有可能的整数时，我们至少能找到一个 k 值使得该语句成立。

为什么以上三个例子被认为是命题，而我们之前却认为“x + 3x = 5”不是命题？原因在于，在这些例子中，存在一个我们所处的“宇宙”，而这些陈述是在这个宇宙上进行量化的。为了用数学方式表达这些陈述，我们需要两个量词：全称量词 ∀（“对于所有”）和存在量词 ∃（“存在”）。例如：

（1）“有些哺乳动物会产卵。”从数学角度来说，“有些”意味着“至少有一个”，所以这句话的意思是“存在一个哺乳动物 x，使得 x 产卵。”如果我们把集合 U 定义为所有哺乳动物的集合，那么我们可以写成：(∃x ∈ U)(x 产卵)。（有时，当集合很明确时，我们可以省略 U，直接写成 ∃x(x 产卵)。）

（2）“对于所有自然数 n，n+ n + 41 是素数”，可以通过将我们的宇宙视为自然数集，记为 N 来表达：（∀n ∈ N）（“n+ n + 41 是素数）。

当用某个值替换变量后，该语句的真假关系变为真时，我们就称该语句为谓词或命题公式。例如，“n + n + 41 是质数”就是一个谓词或命题公式，其中变量为 n。该语句关于自然数 n 的值取决于 n 的值，其真假取决于 n 的值。也就是说，用某个值替换变量后，谓词就变成了命题。具体来说，当 n = 1 时，“43 是质数”为真。而当 n = 41 时，“41 + 41 + 41 是质数”为假，因为 41 + 41 + 41 可以分解为 41 × 43。

注意，在有限论域中，我们可以不用量词，分别用析取和合取来表达存在量词和全称量词命题。例如，如果我们的论域 U 是 {1, 2, 3, 4}，那么 (∃x ∈ U)P(x) 逻辑等价于 P(1) ∨ P(2) ∨ P(3) ∨ P(4)，并且 (∀x ∈ U)P(x) 逻辑等价于 P(1) ∧ P(2) ∧ P(3) ∧ P(4)。然而，在无限论域（例如自然数域）中，这是不可能的。

概念检查！使用量词表达以下两个陈述：“对于所有整数 x，2x + 1 是奇数”，以及“存在一个介于 2 和 4 之间的整数”。

有些语句可以包含多个量词。然而，正如我们将看到的，量词之间并不存在交换律。你只需思考一下英语语句就能明白这一点。请看下面这个（相当血腥的）例子：

> “每次我在纽约坐地铁，都会有人被刺伤。”
>
> “每次我在纽约乘坐地铁，都会发生一起刺伤事件，而这个人似乎就是罪魁祸首。”

第一句话的意思是，每次我坐地铁都会有人被刺伤，但每次可能是不同的人。第二句话则描述了一件非常可怕的事情：有个叫乔的倒霉蛋，每次我坐纽约地铁，都会看到他被刺伤。（可怜的乔一看到我就会拼命逃命。）

从数学角度来看，我们是在两个宇宙上进行量化：T = {我乘坐地铁的次数} 和 P = {人数}。第一个陈述可以写成：(∀t ∈ T)(∃p ∈ P)(p 在时间 t 被刺伤)。第二个陈述可以写成：

statement says: $(\exists p \in P)(\forall t \in T)(p$ gets stabbed at time $t)$.

Let's look at a more mathematical example:

Consider

1. $(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})(x < y)$
2. $(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})(x < y)$

The first statement says that, given an integer, I can find a larger one. The second statement says something very different: that there is a largest integer! The first statement is true, the second is not.

3 Much Ado About Negation

What does it mean for a proposition P to be false? It means that its negation $\neg P$ is true. It's helpful to have some rules for working with negation, as will become more obvious next lecture when we look at proofs.

First, let's look at how to negate conjunctions and disjunctions:

$$\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$$

$$\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)$$

These two equivalences are known as *De Morgan's Laws*, and they are quite intuitive: for example, if it is not the case that $P \wedge Q$ is true, then either P or Q must be false (and vice versa).

Concept check! Verify both of De Morgan's Laws by writing down the appropriate truth tables.

Negating propositions involving quantifiers actually follows analogous laws. Let's start with a simple example. Assume that the universe is $\{1,2,3,4\}$ and let $P(x)$ denote the propositional formula "$x^2 > 10$." Check that $\exists x P(x)$ is true but $\forall x P(x)$ is false. Observe that both $\neg(\forall x P(x))$ and $\exists x \neg P(x)$ are true because $P(1)$ is false. Also note that both $\forall x \neg P(x)$ and $\neg(\exists x P(x))$ are false, since $P(4)$ is true. The fact that each pair of statements had the same truth value is no accident, as the equivalences

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

are laws that hold for any proposition P quantified over any universe (including infinite ones).

It is helpful to think of English sentences to convince yourself (informally) that these laws are true. For example, assume that we are working within the universe $\mathbb{Z}$ (the set of all integers), and that $P(x)$ is the proposition "x is odd." We know that the statement $(\forall x P(x))$ is false, since not every integer is odd. Therefore, we expect its negation, $\neg(\forall x P(x))$, to be true. But how would you say the negation in English? Well, if it is not true that every integer is odd, then there must exist some integer which is not odd (i.e., even). How would this be written in propositional form? That's easy, it's just: $(\exists x \neg P(x))$.

To see a more complex example, fix some universe and propositional formula $P(x,y)$. Assume we have the proposition $\neg(\forall x \exists y P(x,y))$ and we want to push the negation operator inside the quantifiers. By the above laws, we can do it as follows:

$$\neg(\forall x \exists y P(x,y)) \equiv \exists x \neg(\exists y P(x,y)) \equiv \exists x \forall y \neg P(x,y).$$

Notice that we broke the complex negation into a smaller, easier problem as the negation propagated itself through the quantifiers. Note also that the quantifiers "flip" as we go.

声明如下：$(\exists p \in P)(\forall t \in T)$(p 在时间 t 被刺伤)。

让我们来看一个更具数学意义的例子：

考虑

1. $(\forall x \in Z)(\exists y \in Z)(x < y)$
2. $(\exists y \in Z)(\forall x \in Z)(x < y)$

第一句话说，给定一个整数，我可以找到一个更大的整数。第二句话则完全不同：它说存在一个最大的整数！第一句话是正确的，第二句话是错误的。

3. 否定之风

命题 P 为假意味着什么？这意味着它的否定 ¬P 为真。掌握一些处理否定的规则很有帮助，这一点在下一节课我们学习证明时会更加明显。

首先，我们来看如何否定合取式和析取式：

$$\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$$

$$\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)$$

这两个等价关系被称为德摩根定律，它们非常直观：例如，如果 $P \wedge Q$ 不为真，那么 P 或 Q 必定为假（反之亦然）。

概念检查！通过编写相应的真值表来验证德摩根定律的两条定律。

否定包含量词的命题实际上遵循类似的规律。我们先来看一个简单的例子。假设全集为 {1, 2, 3, 4}，令 P(x) 表示命题公式“x > 10”。验证 $\exists x P(x)$ 为真，但 $\forall x P(x)$ 为假。注意到 $\neg(\forall x P(x))$ 和 $\exists x \neg P(x)$ 都为真，因为 P(1) 为假。还要注意，$\forall x \neg P(x)$ 和 $\neg(\exists x P(x))$ 都为假，因为 P(4) 为真。每对语句的真值相同并非偶然，因为等价关系……

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

是适用于任何命题 P 在任何宇宙（包括无限宇宙）上量化的定律。

为了（非正式地）说服自己这些定律是正确的，思考一些英文句子会很有帮助。例如，假设我们是在集合 Z（所有整数的集合）中讨论，并且 P(x) 是命题“x 是奇数”。我们知道命题 $(\forall x P(x))$ 是假的，因为并非每个整数都是奇数。因此，我们预期它的否定 $\neg(\forall x P(x))$ 为真。但是，如何用英语表达这个否定呢？如果并非每个整数都是奇数，那么必然存在某个整数不是奇数（即偶数）。这用命题形式如何表达呢？很简单，只需：$(\exists x \neg P(x))$。

为了看一个更复杂的例子，我们固定一个论域和命题公式 P(x, y)。假设我们有命题 $\neg(\forall x \exists y P(x, y))$，并且我们想把否定运算符放在量词内部。根据上述法则，我们可以这样做：

$$\exists x \exists y P(x,y)) \; \exists x \exists y P(x,y)) \; \exists x \forall y P(x,y)。$$

请注意，随着否定关系在量词间的传递，我们将复杂的否定问题分解成了一个更小、更容易解决的问题。还要注意的是，量词会随着运算顺序而“翻转”。

4 Trickier Quantifier Examples

Let's look at a trickier set of examples:

Write the sentence "there are **at least** three distinct integers x that satisfy $P(x)$" as a proposition using quantifiers! One way to do it is

$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge z \neq x \wedge P(x) \wedge P(y) \wedge P(z)).$$

(Here all quantifiers are over the universe $\mathbb{Z}$ of integers.)

Now write the sentence "there are **at most** three distinct integers x that satisfy $P(x)$" as a proposition using quantifiers. One way to do it is

$$\exists x \exists y \exists z \forall d (P(d) \implies d = x \vee d = y \vee d = z).$$

If you're not sure how to interpret this propositional logic statement, you can read it as: "There exist 3 three specific integers x, y, and z, such that if $P(d)$ is true, then d has to be one of those three integers." There are many other ways to express this same statement, for example:

$$\forall x \forall y \forall v \forall z ((x \neq y \wedge y \neq v \wedge v \neq x \wedge x \neq z \wedge y \neq z \wedge v \neq z) \implies \neg(P(x) \wedge P(y) \wedge P(v) \wedge P(z))).$$

Concept check! Check that you understand both of the above alternatives.

Note that we can also generate even more exactly identical alternatives, e.g. writing the contrapositives, propagating negations, etc. In general, you want to pick the propositional logic statement that most clearly expresses the idea.

Finally, what if we want to express the sentence "there are **exactly** three distinct integers x that satisfy $P(x)$"? This is now easy: we can just use the *conjunction* of the two propositions above.

4 个更棘手的量词示例

让我们来看一组更棘手的例子：

请用量词将句子“至少有三个不同的整数 x 满足 P(x)”写成命题！一种方法是：

$$\exists x \exists y \exists z (x \neq y \land y \neq z \land z \neq x \land P(x) \land P(y) \land P(z))。$$

（这里所有量词都针对整数集合 Z。）

现在，请用量词将句子“至多有三个不同的整数 x 满足 P(x)”写成一个命题。一种方法是：

$$\exists x \exists y \exists z \forall d (P(d) \Longrightarrow d = x \lor d = y \lor d = z)。$$

如果你不确定如何理解这个命题逻辑语句，你可以把它理解为： “存在三个特定的整数 x、y 和 z，使得如果 P(d) 为真，那么 d 必须是这三个整数中的一个。” 表达这个命题还有许多其他方式，例如：

$$\forall x \forall y \forall v \forall z ((x \neq y \land y \neq v \land v \neq x \land x \neq z \land y \neq z \land v \neq z) \Longrightarrow \neg(P(x) \land P(y) \land P(v) \land P(z))).$$

概念检查！请确认您理解以上两种方案。

请注意，我们还可以生成更多完全相同的替代方案，例如编写逆否命题、传播否定等。一般来说，您需要选择最清晰表达该思想的命题逻辑语句。

最后，如果我们想表达“恰好有三个不同的整数 x 满足 P(x)”这个命题呢？这就很容易了：我们只需使用上面两个命题的合取即可。